Channels

SWAT+ Documentation

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Purpose

This page documents the legacy channel input files: `channel.cha` (original reach pointer), `channel-lte.cha` (SWAT+ deg routing pointer), `hyd-sed-lte.cha` (combined hydrology and sediment table used by the deg routine), `initial.cha` (initial-state pointer), and `temperature.cha` (water temperature parameters). All file names are set in `type input_cha` in `src/input_file_module.f90`. Each file is independently optional: if the slot in `file.cio` is `null` or the file is missing, the reader allocates an empty array and skips on.

The three companion tables that these pointer files reference (`hydrology.cha`, `sediment.cha`, `nutrients.cha`) are documented on the sd_channel page.

Source

- File names: src/input_file_module.f90, `type input_cha`
- Readers:
  - src/ch_read.f90 (`channel.cha`)
  - src/sd_channel_read.f90 (`channel-lte.cha`)
  - src/sd_hydsed_read.f90 (`hyd-sed-lte.cha`)
  - src/ch_read_init.f90 (`initial.cha`)
  - src/ch_read_temp.f90 (`temperature.cha`)
- Type definitions: src/channel_data_module.f90, src/sd_channel_module.f90

Every reader skips two header lines, then reads one record per channel definition.

channel.cha

Pointer file for the original SWAT-style reach routing. Each record names a channel and points to entries in `initial.cha`, `hydrology.cha`, `sediment.cha`, and `nutrients.cha`. Read into `ch_dat_c` (`type channel_data_char_input`).

#	Column	Type	Description
1	id	int	record number (echoed back)
2	name	char(16)	channel name
3	init	char(16)	name of an `initial.cha` entry, or `null`
4	hyd	char(16)	name of a `hydrology.cha` entry, or `null`
5	sed	char(16)	name of a `sediment.cha` entry, or `null`
6	nut	char(16)	name of a `nutrients.cha` entry, or `null`

After reading, the reader resolves each character pointer to an integer index by scanning `ch_init`, `ch_hyd`, `ch_sed`, and `ch_nut`. Missing names are written to the diagnostics file (unit 9001) with messages like `"<name> not found (hydrology.cha)"`. The reader does not abort; the index is left at 0 and downstream code uses the default record.

No example dataset shipping with SWAT+ uses `channel.cha`. The two reference datasets at `refdata/Osu_1hru` and (if present) `refdata/Ames_sub1` configure this slot as `null` in `file.cio` and use `channel-lte.cha` instead.

channel-lte.cha

Pointer file for SWAT+ deg routing. Listed under `chandeg.con`. Each record points to entries in `initial.cha`, `hyd-sed-lte.cha`, an optional sediment table, and `nutrients.cha`. Read into `sd_dat` (`type swatdeg_datafiles`).

#	Column	Type	Description
1	id	int	record number used to position the array element
2	name	char(16)	channel name
3	cha_ini	char(16)	name of an `initial.cha` entry, or `null`
4	cha_hyd	char(16)	name of a `hyd-sed-lte.cha` entry, or `null`
5	cha_sed	char(16)	name of a sediment-nutrient entry (resolved against `sed_nut.cha` when present)
6	cha_nut	char(16)	name of a `nutrients.cha` entry, or `null`

The `cha_hyd` column is matched against names in both `sd_chd` (`hyd-sed-lte.cha`) and `sd_chd1` (`sed_nut.cha`). The `cha_sed` column is read but not resolved to an index by `sd_channel_read`; sediment behaviour follows the `hyd-sed-lte.cha` parameters.

Example from `refdata/Osu_1hru/channel-lte.cha`:

```
channel-lte.cha: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
      id  name                 cha_ini           cha_hyd           cha_sed           cha_nut
       1  cha01                initcha1          hydcha01             null           nutcha1
       2  cha03                initcha1          hydcha03             null           nutcha1
       3  cha04                initcha1          hydcha04             null           nutcha1
```

hyd-sed-lte.cha

Combined hydrology and bank-erosion parameters for the SWAT+ deg routing routine. One record per channel hydrology profile. Read into `sd_chd` (`type swatdeg_hydsed_data`).

#	Column	Type	Units	Default	Description
1	name	char(25)	none	“”	parameter set name (referenced by `channel-lte.cha`)
2	order	int	none	0	stream order
3	chw	real	m	0.0	channel width
4	chd	real	m	0.0	channel depth
5	chs	real	m/m	0.0	channel slope
6	chl	real	km	0.0	channel length
7	chn	real	none	0.0	channel Manning’s n
8	chk	real	mm/hr	0.0	channel bottom hydraulic conductivity
9	bank_exp	real	none	0.0	bank erosion exponent (column `erod_fact` in editor output)
10	cov	real	0..1	0.0	channel cover factor
11	sinu	real	none	0.0	sinuosity (channel length / straight-line length)
12	vcr_coef	real	none	0.0	critical velocity coefficient (column `eq_slp` in editor output)
13	d50	real	mm	0.0	median sediment diameter
14	ch_clay	real	percent	0.0	clay percent of bank and bed
15	carbon	real	percent	0.0	carbon percent of bank and bed
16	ch_bd	real	t/m^3	0.0	dry bulk density
17	chss	real	none	0.0	channel side slope
18	bankfull_flo	real	m^3/s	0.0	bank-full flow rate (column `bed_load` in editor output)
19	fps	real	m/m	0.000001	flood plain slope
20	fpn	real	none	0.1	flood plain Manning’s n
21	n_conc	real	mg/kg	0.0	nitrogen concentration in channel bank
22	p_conc	real	mg/kg	0.0	phosphorus concentration in channel bank
23	p_bio	real	fraction	0.0	fraction of bank P that is bioavailable

Example from `refdata/Osu_1hru/hyd-sed-lte.cha`:

```
hyd-sed-lte.cha: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
name                          order            wd            dp           slp           len          mann
hydcha01                          4      27.17726       0.99164       0.00146       3.42760       0.05000
```

Editor column names differ from source field names. The reader uses position only.

initial.cha

Initial state pointer file. Each record names a set of initial-condition files for organic-mineral mass, pesticides, pathogens, heavy metals, and salts. Read into `ch_init` (`type channel_init_datafiles`).

#	Column	Type	Description
1	name	char(16)	initial-state set name
2	org_min	char(16)	initial organic-mineral input file name, or `null`
3	pest	char(16)	initial pesticide input file name, or `null`
4	path	char(16)	initial pathogen input file name, or `null`
5	hmet	char(16)	initial heavy metals input file name, or `null`
6	salt	char(16)	initial salt input file name, or `null`

The reader uses unformatted list-directed reads with no leading integer id column. The record is read directly into the derived type, so the first token on each line is the `name`.

Example from `refdata/Osu_1hru/initial.cha`:

```
initial.cha: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
name                        org_min              pest              path              hmet              salt
initcha1                    no_init              null              null              null              null
```

A separate file `initial.cha_cs` is read by `ch_read_init_cs` for salt and conservative-species initial conditions. The file name is hard-coded in the reader (not in `type input_cha`).

temperature.cha

Water temperature parameters used by the channel temperature routine. Read into `w_temp` (`type water_temperature_data`). The `name` field is `character(len=13)` in the source; longer names will be truncated.

#	Column	Type	Units	Default	Description
1	name	char(13)	none	“”	parameter set name
2	sno_mlt	real	none	1.0	coefficient on snowmelt temperature contribution
3	gw	real	none	1.0	coefficient on groundwater temperature contribution
4	sur_lat	real	none	1.0	coefficient on surface and lateral flow temperature contribution
5	sno_lag	real	days	2.0	air temperature lag to snowmelt (1..3)
6	gw_lag	real	days	250.0	air temperature lag to groundwater flow (200..365)
7	surf_lag	real	days	5.0	air temperature lag to surface runoff (2..5)
8	lat_lag	real	days	10.0	air temperature lag to lateral flow (5..10)
9	lat_lag_coef	real	none	0.75	lateral air-lag coefficient
10	surf_lag_coef	real	none	0.75	surface air-lag coefficient (also used for snow)
11	gw_lag_coef	real	none	0.75	groundwater air-lag coefficient
12	hex_coef1	real	none	0.75	dew point calibration
13	hex_coef2	real	none	1.5	channel geometry calibration
14	sf_on	int	none	0	shade factor file activation. `1` = read from file, `0` = use value from calibration file
15	ssff	real	0..1	0.5	shade factor value (used when `sf_on = 0`)

No example dataset in `refdata/` contains `temperature.cha`. The slot is `null` in `file.cio` for the included reference datasets.

Related files

- `file.cio`. The `channel` line selects which of these files are read.
- `chandeg.con`. Connectivity file that references `channel-lte.cha` entries by name.
- `channel.con`. Connectivity file that references `channel.cha` entries by name.
- sd_channel companion tables (`hydrology.cha`, `sediment.cha`, `nutrients.cha`).

Purpose

This page documents the three per-channel parameter tables that hold hydraulic, sediment, and nutrient coefficients used inside the channel routing modules: `hydrology.cha`, `sediment.cha`, and `nutrients.cha`. Each is a flat database. Records are referenced by name from the pointer files `channel.cha` (legacy) and `channel-lte.cha` (SWAT+ deg). When the SWAT+ deg routine routes water and constituents through reaches listed under `chandeg.con`, the parameters in these tables drive the in-channel processes.

File names are declared in `type input_cha` in `src/input_file_module.f90` (`hyd`, `sed`, `nut`). Each file is optional. If the slot in `file.cio` is `null` or the file is missing, the reader allocates an empty array and the matching pointer is left at index 0.

Source

- File names: `src/input_file_module.f90`, `type input_cha`
- Readers:
  - `src/ch_read_hyd.f90`
  - `src/ch_read_sed.f90`
  - `src/ch_read_nut.f90`
- Type definitions: `src/channel_data_module.f90`, types `channel_hyd_data`, `channel_sed_data`, `channel_nut_data`

Every reader skips two header lines, then reads one record per parameter set.

hydrology.cha

Hydraulic and bank parameters for the original SWAT-style reach routine. Read into `ch_hyd` (`type channel_hyd_data`). The reader rewrites a few fields after read: `alpha_bnk` is replaced with `exp(-alpha_bnk)`, `s` is floored at 0.0001, `n` is clamped to `[0.01, 0.70]`, `l` is floored at 0.0010, `wdr` defaults to 3.5 if non-positive, and `side` defaults to 2.0 if non-positive.

#	Column	Type	Units	Default	Description
1	name	char(16)	none	“default”	parameter set name
2	w	real	m	2.0	average width of main channel
3	d	real	m	0.5	average depth of main channel
4	s	real	m/m	0.01	average slope of main channel
5	l	real	km	0.1	main channel length in subbasin
6	n	real	none	0.05	Manning’s n
7	k	real	mm/hr	0.01	effective hydraulic conductivity of channel alluvium
8	wdr	real	m/m	6.0	channel width-to-depth ratio
9	alpha_bnk	real	days	0.03	alpha factor for bank storage recession (read as days, stored as `exp(-alpha_bnk)`)
10	side	real	none	0.0	channel side slope (horizontal per unit vertical)

No example dataset in `refdata/` ships with `hydrology.cha`. The included reference datasets configure this slot as `null` in `file.cio` and use `hyd-sed-lte.cha` instead.

sediment.cha

Channel sediment routing parameters. Read into `ch_sed` (`type channel_sed_data`). The 12 monthly `erod` values follow the scalar fields. The reader clamps several values: cover and erodibility factors are bounded by the routing equation choice, bank and bed `d50` default to 50 and 500 micrometers if non-positive, bank and bed bulk density default to 1.40 and 1.50 g/cc if non-positive, and bank and bed jet-test erodibility default to 0.2 (or `0.2 / sqrt(tc_*)` if a critical shear stress is given). If all 12 monthly `erod` values are effectively zero, the reader fills the monthly array with `cov1`.

#	Column	Type	Units	Default	Description
1	name	char(16)	none	“”	parameter set name
2	eqn	int	none	0	sediment routing method. `0` original SWAT, `1` Bagnold, `2` Kodatie, `3` Molinas-Wu, `4` Yang
3	cov1	real	none	0.1	channel erodibility factor (0..1 when `eqn = 0`, 0..25 otherwise)
4	cov2	real	none	0.1	channel cover factor (same bounds as `cov1`)
5	bnk_bd	real	g/cc	0.0	bulk density of channel bank sediment (1.1..1.9)
6	bed_bd	real	g/cc	0.0	bulk density of channel bed sediment (1.1..1.9)
7	bnk_kd	real	cm^3/N/s	0.0	bank erodibility from jet test
8	bed_kd	real	cm^3/N/s	0.0	bed erodibility from jet test
9	bnk_d50	real	micrometers	0.0	bank median particle diameter (clamped to 10000 max)
10	bed_d50	real	micrometers	0.0	bed median particle diameter (clamped to 10000 max)
11	tc_bnk	real	N/m^2	0.0	critical shear stress of channel bank
12	tc_bed	real	N/m^2	0.0	critical shear stress of channel bed
13..24	erod(1..12)	real	none	0.0	monthly channel erodibility factor. `0.0` = non-erosive, `1.0` = no resistance to erosion

No example dataset in `refdata/` ships with `sediment.cha`.

nutrients.cha

In-stream nutrient cycling parameters. Read into `ch_nut` (`type channel_nut_data`). After read, several rate constants are scaled by the routing time step (`time%step`) so that values defined per day are converted to per-substep. The light half-saturation `k_l` is converted from kJ/(m^2 min) to MJ/(m^2 hr) by multiplying by 0.06. Defaults are applied (in place of zero) for every coefficient.

#	Column	Type	Units	Default	Description
1	name	char(16)	none	“”	parameter set name
2	onco	real	ppm	0.0	channel organic N concentration
3	opco	real	ppm	0.0	channel organic P concentration
4	rs1	real	m/day	1.0	local algal settling rate at 20 deg C
5	rs2	real	(mg disP-P)/(m^2 day)	0.05	benthos source rate for dissolved P at 20 deg C
6	rs3	real	(mg NH4-N)/(m^2 day)	0.5	benthos source rate for ammonia at 20 deg C
7	rs4	real	1/day	0.05	organic N settling rate at 20 deg C
8	rs5	real	1/day	0.05	organic P settling rate at 20 deg C
9	rs6	real	1/day	2.5	settling rate for arbitrary non-conservative constituent
10	rs7	real	(mg ANC)/(m^2 day)	2.5	benthal source rate for arbitrary non-conservative constituent
11	rk1	real	1/day	1.71	CBOD deoxygenation rate at 20 deg C
12	rk2	real	1/day	1.0	reaeration rate (Fickian) at 20 deg C
13	rk3	real	1/day	2.0	CBOD loss rate due to settling at 20 deg C
14	rk4	real	(mg O2)/(m^2 day)	0.0	sediment oxygen demand at 20 deg C
15	rk5	real	1/day	1.71	coliform die-off rate
16	rk6	real	1/day	1.71	decay rate for arbitrary non-conservative constituent
17	bc1	real	1/hr	0.55	NH3 to NO2 biological oxidation rate at 20 deg C
18	bc2	real	1/hr	1.1	NO2 to NO3 biological oxidation rate at 20 deg C
19	bc3	real	1/hr	0.21	organic N to ammonia hydrolysis rate at 20 deg C
20	bc4	real	1/hr	0.35	organic P to dissolved P decay rate at 20 deg C
21	lao	real	none	2.0	QUAL2E light averaging option. Only option 2 is implemented in SWAT
22	igropt	int	none	2	algal growth limiting equation. 1 multiplicative, 2 limiting nutrient, 3 harmonic mean
23	ai0	real	ug chla / mg alg	50.0	ratio of chlorophyll-a to algal biomass
24	ai1	real	mg N / mg alg	0.08	nitrogen fraction of algal biomass
25	ai2	real	mg P / mg alg	0.015	phosphorus fraction of algal biomass
26	ai3	real	mg O2 / mg alg	1.60	rate of O2 production per unit of algal photosynthesis
27	ai4	real	mg O2 / mg alg	2.0	rate of O2 uptake per unit of algal respiration
28	ai5	real	mg O2 / mg N	3.5	rate of O2 uptake per unit of NH3 oxidation
29	ai6	real	mg O2 / mg N	1.07	rate of O2 uptake per unit of NO2 oxidation
30	mumax	real	1/hr	2.0	maximum specific algal growth rate at 20 deg C
31	rhoq	real	1/day	2.5	algal respiration rate
32	tfact	real	none	0.3	fraction of solar radiation that is photosynthetically active
33	k_l	real	kJ/(m^2 min) read, stored as MJ/(m^2 hr)	0.75	light half-saturation coefficient
34	k_n	real	mg N / L	0.02	Michaelis-Menten N half-saturation constant

#	Column	Type	Units	Default	Description
35	k_p	real	mg P / L	0.025	Michaelis-Menten P half-saturation constant
36	lambda0	real	1/m	1.0	non-algal portion of the light extinction coefficient
37	lambda1	real	1/(m * ug chla/L)	0.03	linear algal self-shading coefficient
38	lambda2	real	(1/m)(ug chla/L)^(-2/3)	0.054	nonlinear algal self-shading coefficient
39	p_n	real	none	0.5	algal preference factor for ammonia

Example from `refdata/Osu_1hru/nutrients.cha`:

```
nutrients.cha: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
name                    plt_n         ptl_p       alg_stl      ben_disp      ben_nh3n      ptln_stl      
nutcha1               0.00000       0.00000       1.00000       0.05000       0.50000       0.05000
```

Editor column names (`plt_n`, `ptl_p`, `alg_stl`, …) do not match the source field names (`onco`, `opco`, `rs1`, …). The reader uses position only.

Notes on the Osu_1hru example:

- Position 11 in this dataset (`cbn_bod_co = 1.71`) is `rk1` in the source.
- Position 13 (`cbn_bod_stl = 0.36`) is `rk3`. The source default is 2.0; this dataset overrides it.
- Position 14 (`ben_bod = 2.0`) is `rk4`. The source default is 0.0; this dataset overrides it.
- Positions 21 and 22 (`q2e_lt = 2`, `q2e_alg = 2`) are `lao` (read as real) and `igropt` (read as int).

Related files

- `*.cha` legacy channel files (`channel.cha`, `channel-lte.cha`, `hyd-sed-lte.cha`, `initial.cha`, `temperature.cha`). The pointer files that select records from `hydrology.cha`, `sediment.cha`, and `nutrients.cha` by name.
- `chandeg.con`. Connectivity file for SWAT+ deg routing.
- `channel.con`. Connectivity file for the original SWAT-style routing.
- `file.cio`. The `channel` line lists the file names that occupy each slot in `type input_cha`.